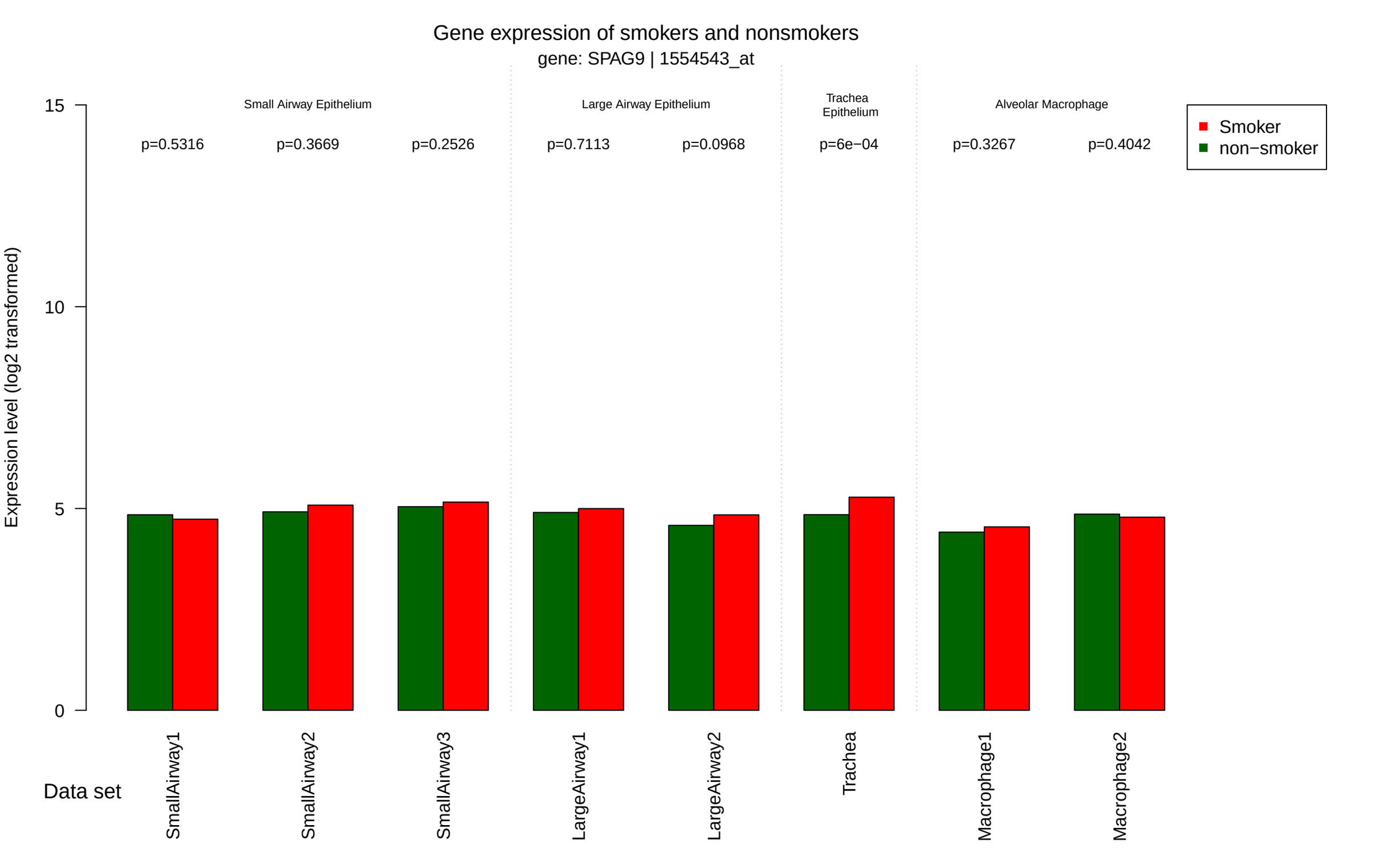
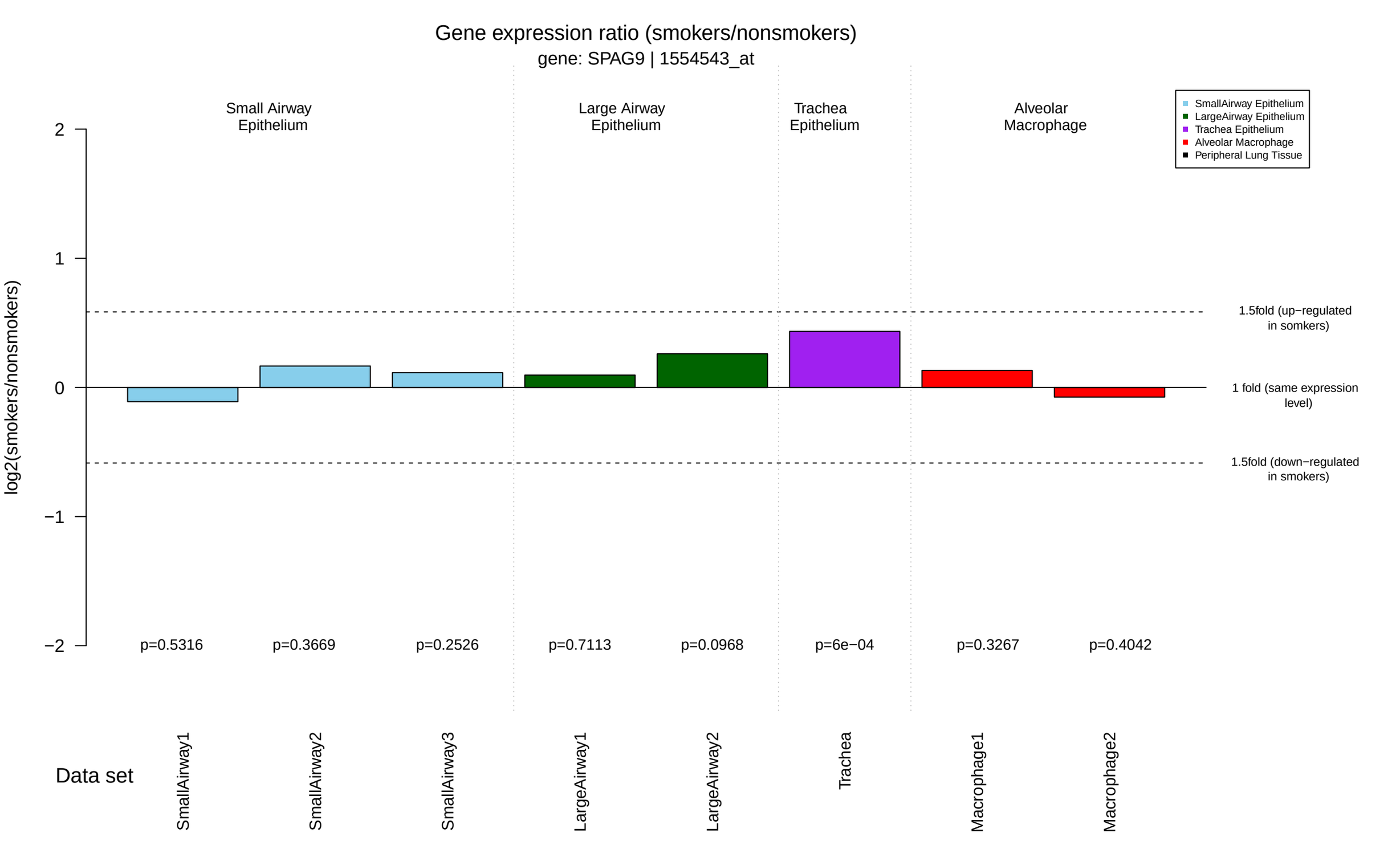
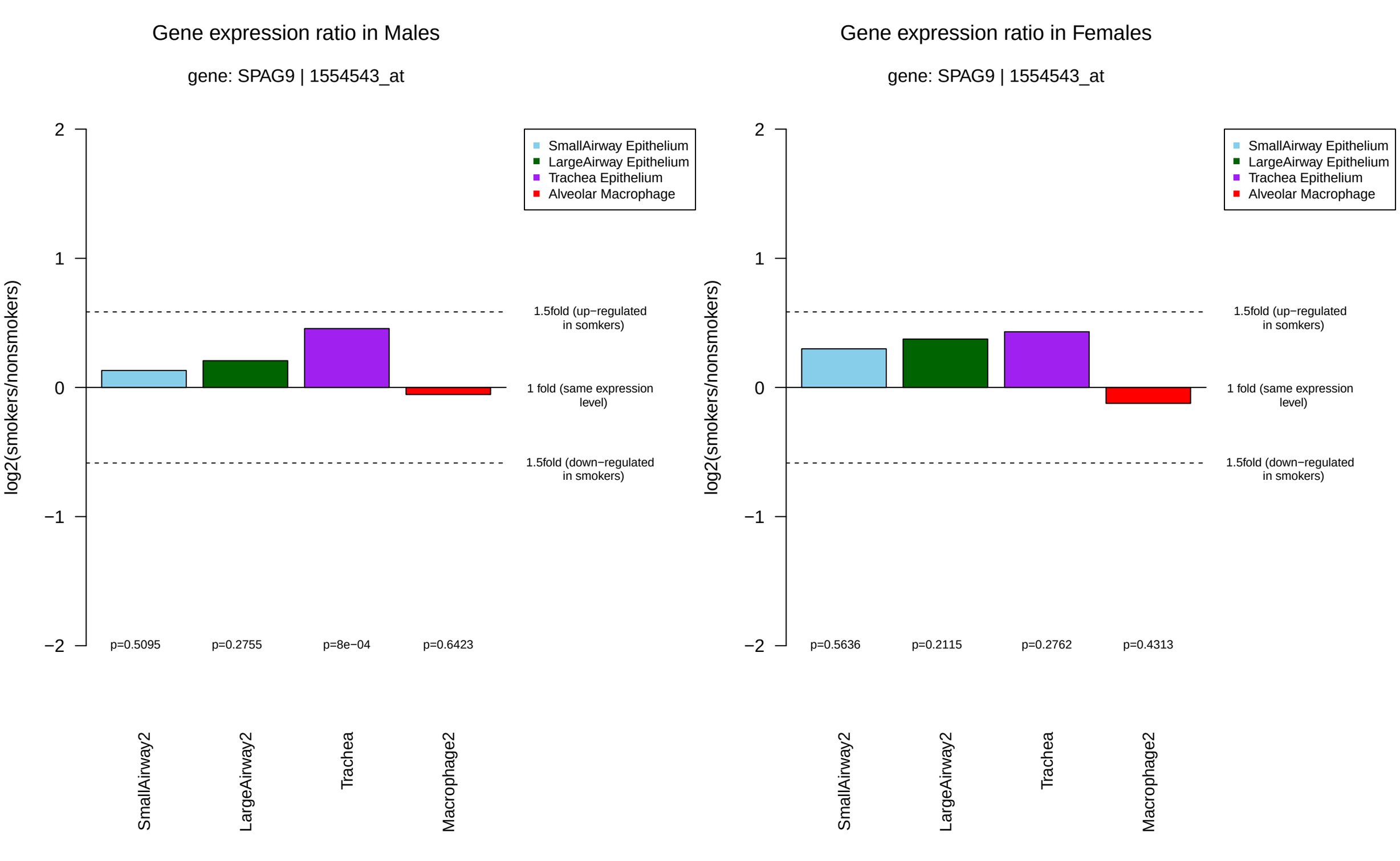
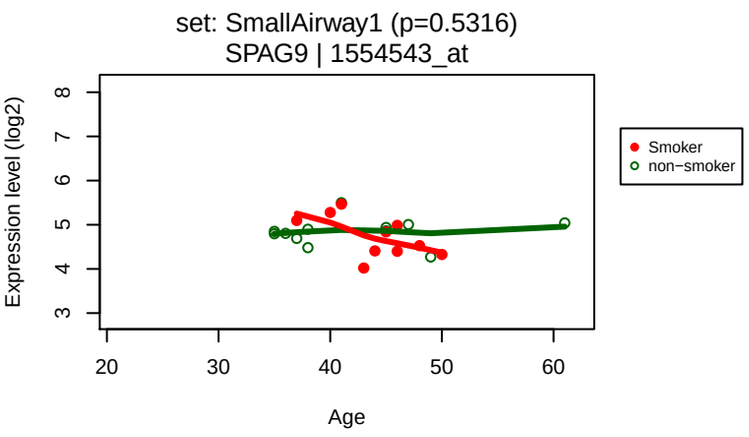
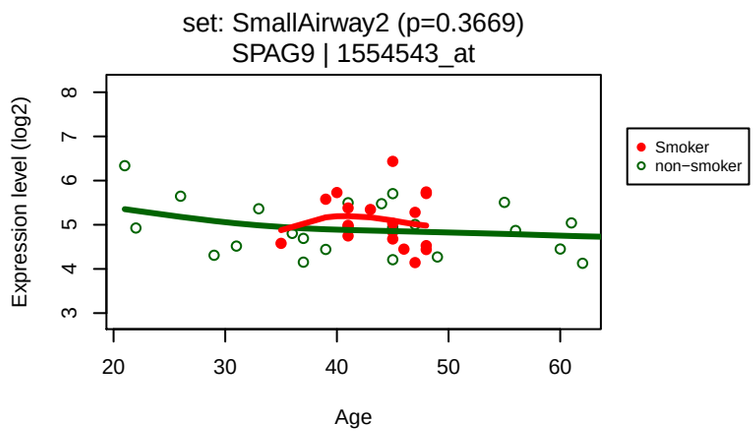
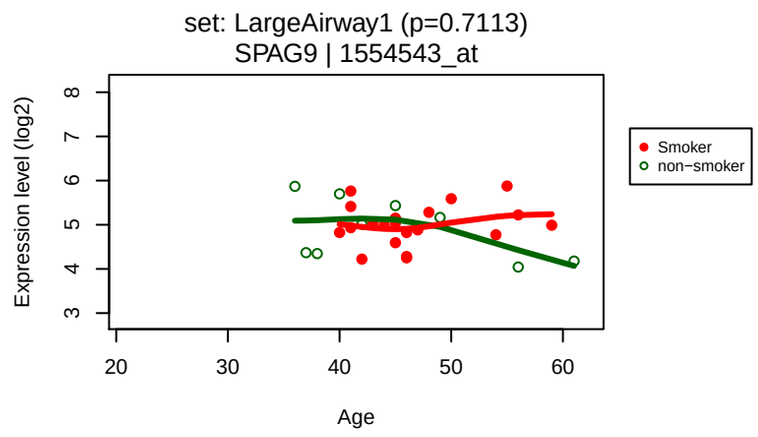

Gene expression of smokers and nonsmokers
gene: SPAG9 | 1554543_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5316
p=0.3669
p=0.2526
p=0.7113
p=0.0968
p=6e−04
p=0.3267
p=0.4042
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

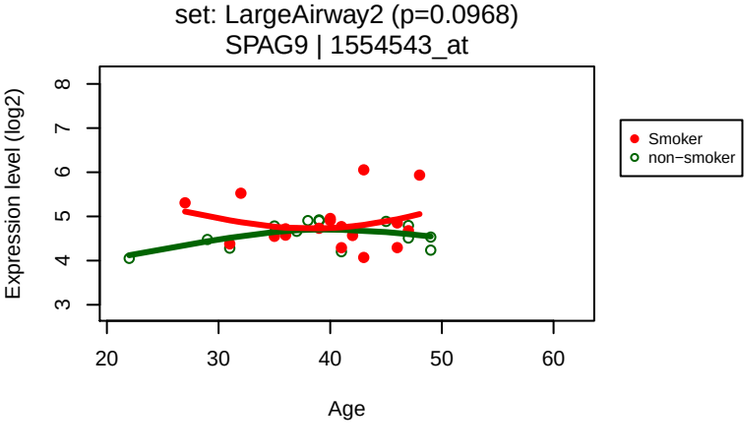

Gene expression ratio (smokers/nonsmokers)
gene: SPAG9 | 1554543_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5316
p=0.3669
p=0.2526
p=0.7113
p=0.0968
p=6e−04
p=0.3267
p=0.4042
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

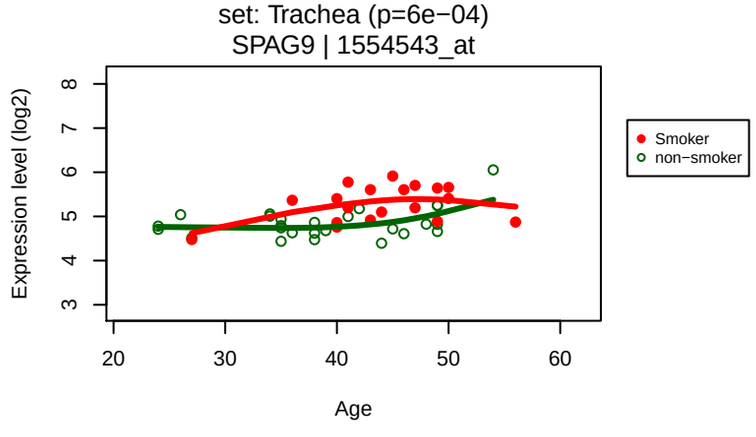

Gene expression ratio in Males
gene: SPAG9 | 1554543_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5095
p=0.2755
p=8e−04
p=0.6423
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPAG9 | 1554543_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5636
p=0.2115
p=0.2762
p=0.4313
SmallAirway2
LargeAirway2
Trachea
Macrophage2

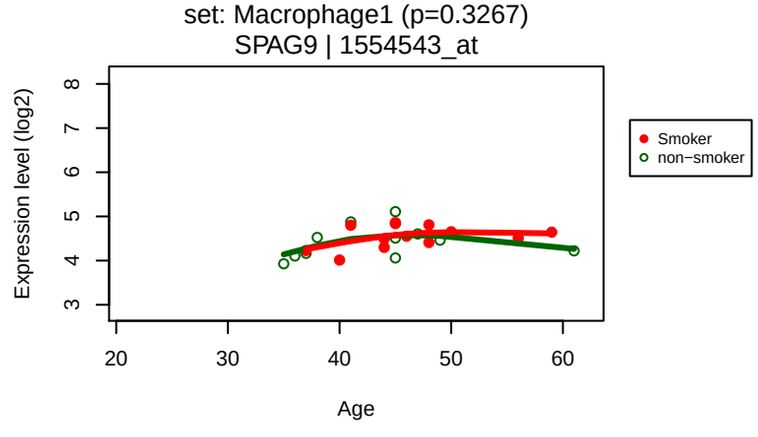

set: SmallAirway1 (p=0.5316)
SPAG9 | 1554543_at
Expression level (log2)
Age
Smoker
non−smoker

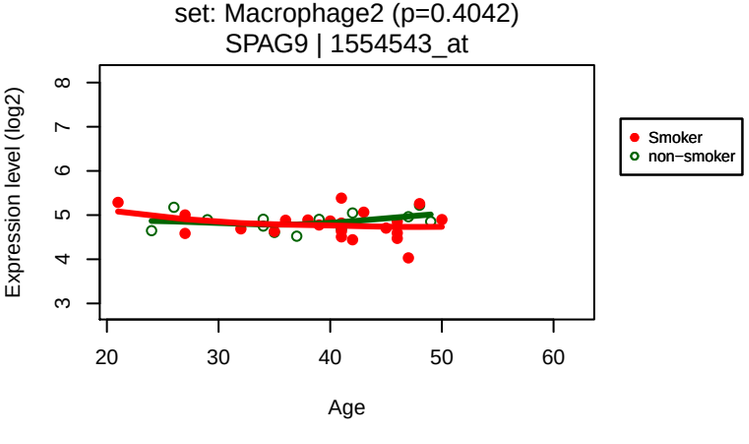

set: SmallAirway2 (p=0.3669)
SPAG9 | 1554543_at
Expression level (log2)
Age
Smoker
non−smoker

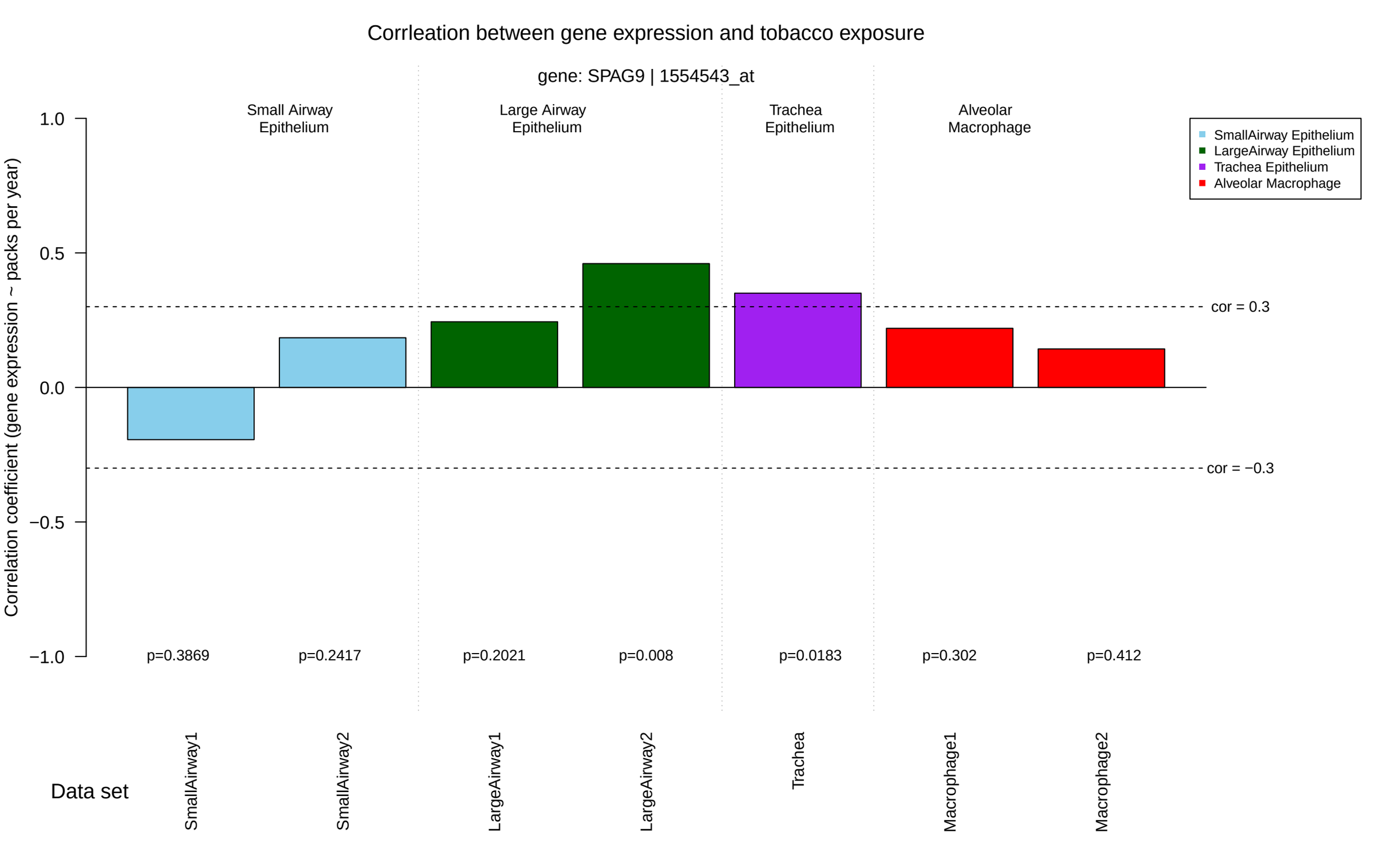

set: LargeAirway1 (p=0.7113)
SPAG9 | 1554543_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0968)
SPAG9 | 1554543_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=6e−04)
SPAG9 | 1554543_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3267)
SPAG9 | 1554543_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4042)
SPAG9 | 1554543_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPAG9 | 1554543_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3869
p=0.2417
p=0.2021
p=0.008
p=0.0183
p=0.302
p=0.412
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2